

## Exp 8

### Aim

To configure a cryptocurrency wallet using MetaMask and integrate it with a decentralized application (DApp) to perform and verify blockchain transactions.

### Theory

**MetaMask** is a cryptocurrency wallet that provides users with an interface to interact with blockchain networks and decentralized applications. It manages blockchain accounts and allows users to approve or reject transactions initiated by DApps.

A **Decentralized Application (DApp)** is an application that interacts with blockchain-based smart contracts. The frontend of a DApp communicates with the blockchain through a Web3 provider such as MetaMask. MetaMask acts as a bridge between the user's browser and the blockchain network.

In this experiment, the **ZeroTrace DApp** was used to demonstrate MetaMask integration. The DApp uses Web3 technologies to connect the user's wallet and interact with deployed smart contracts. After connecting MetaMask, a blockchain transaction was initiated through the DApp. MetaMask displayed the transaction request, which was confirmed by the user. The resulting transaction was then verified using its transaction ID on a blockchain explorer.

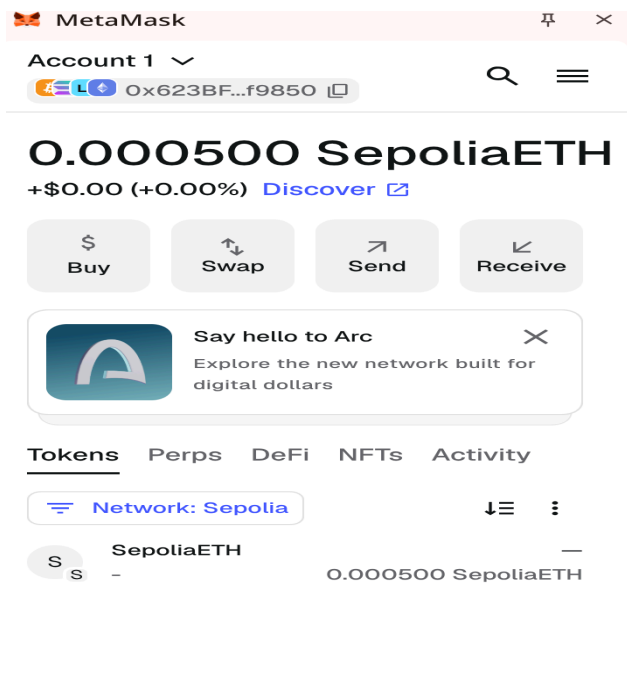
The basic interaction can be represented as:



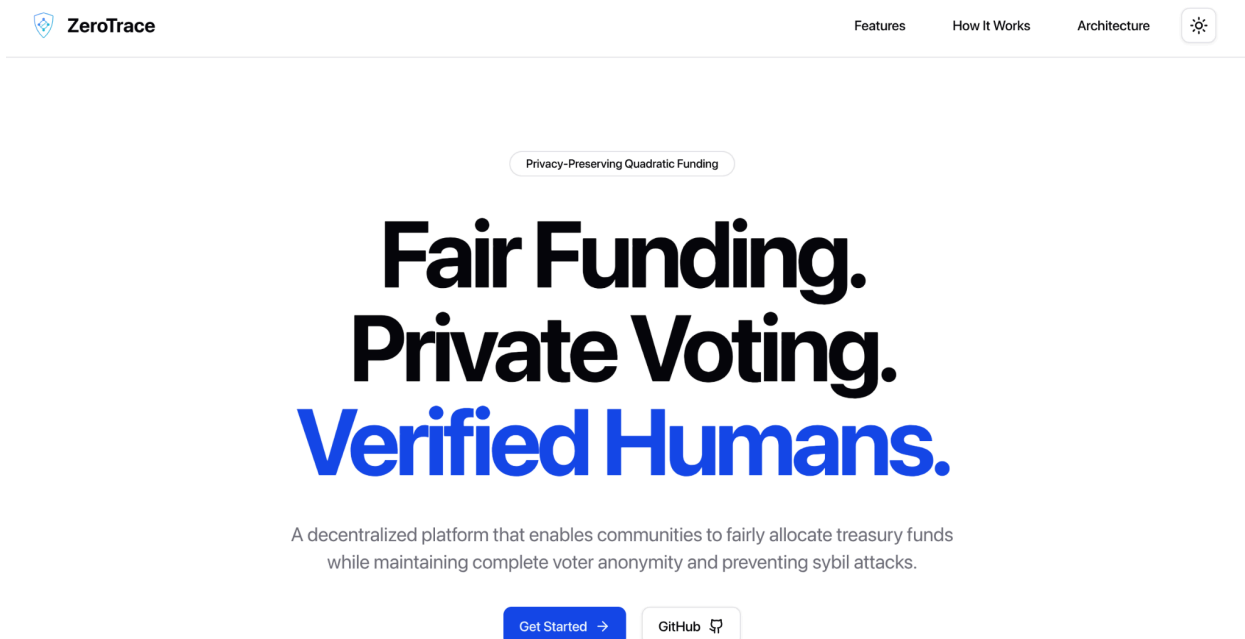
MetaMask provides important functions such as account management, network selection, transaction signing, and connection authorization. A transaction hash is generated for a submitted blockchain transaction and can be used to verify its status and details on a blockchain explorer.

## Output / Screenshot Description

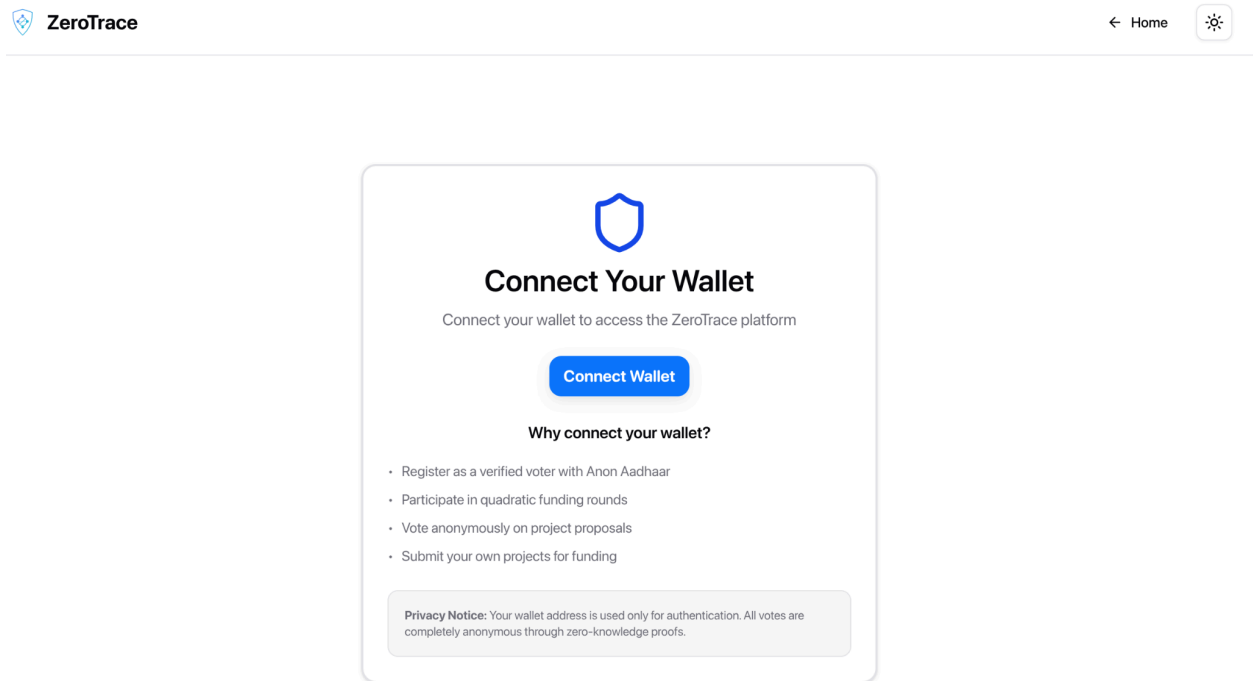
### Screenshot 1: MetaMask Wallet Account



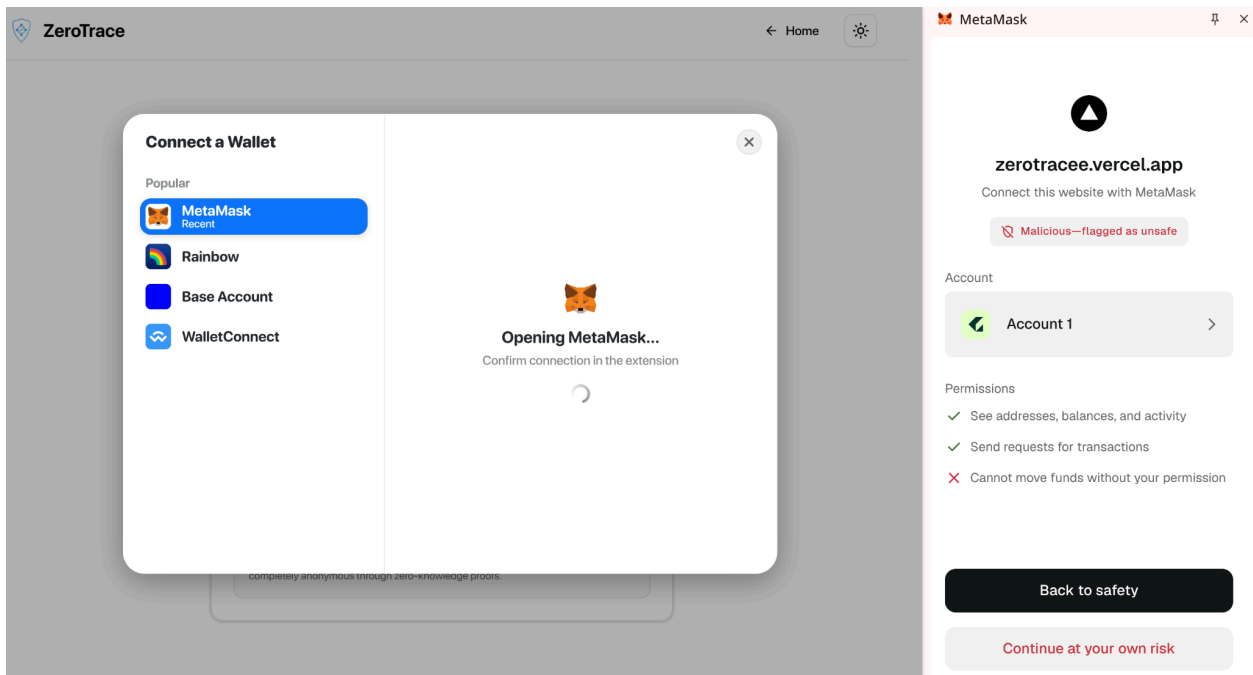
### Screenshot 2: ZeroTrace DApp



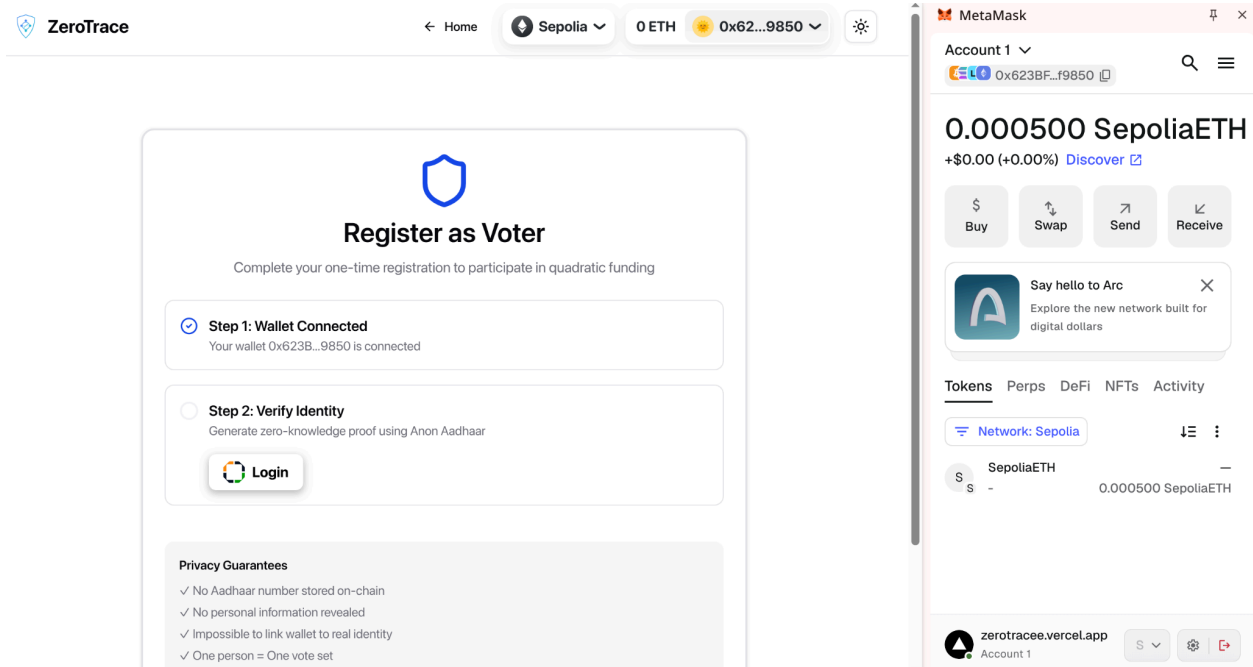
### Screenshot 3: Connect Wallet Option



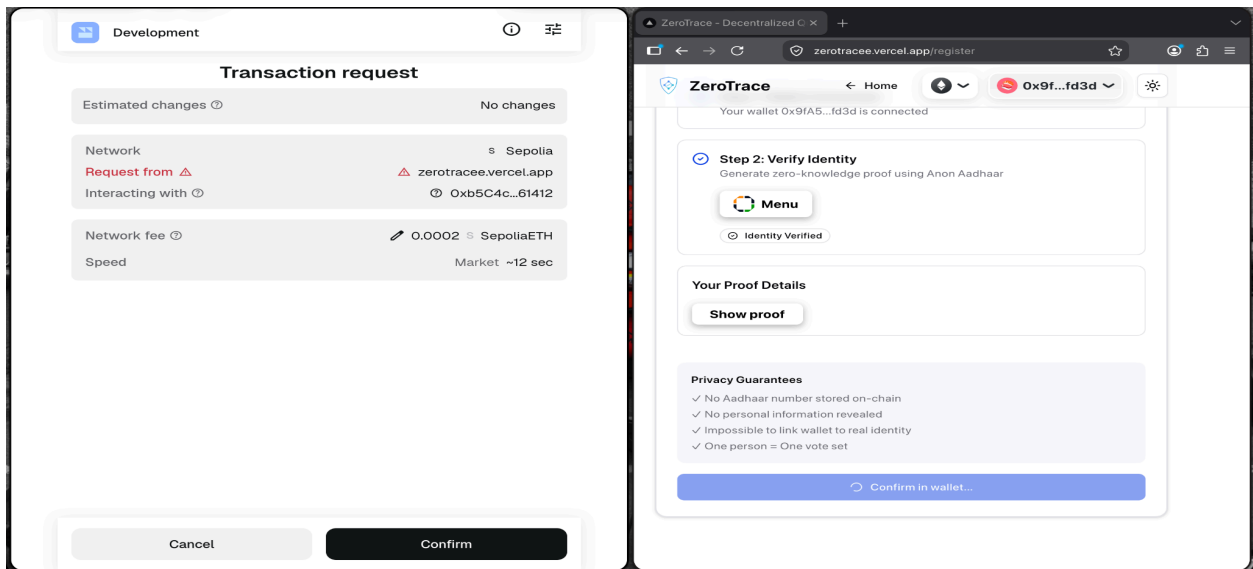
### Screenshot 4: MetaMask Connection Approval

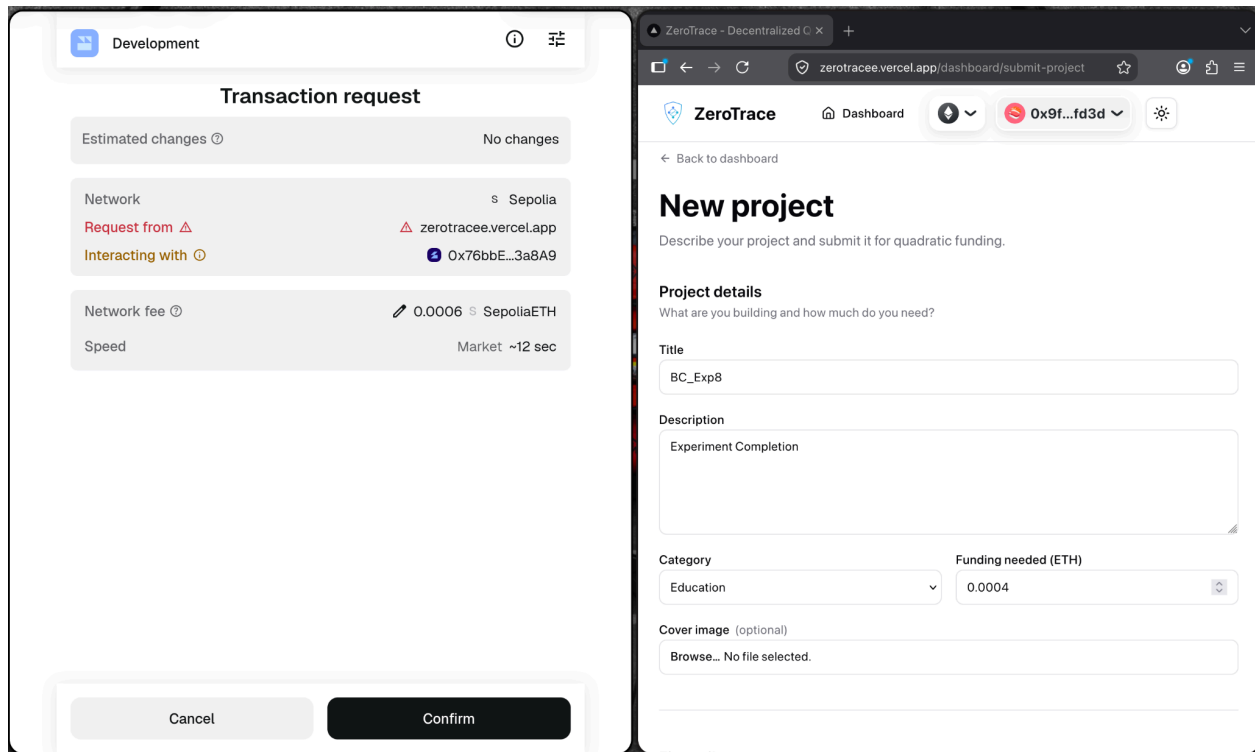


## Screenshot 5: Successfully Connected Wallet

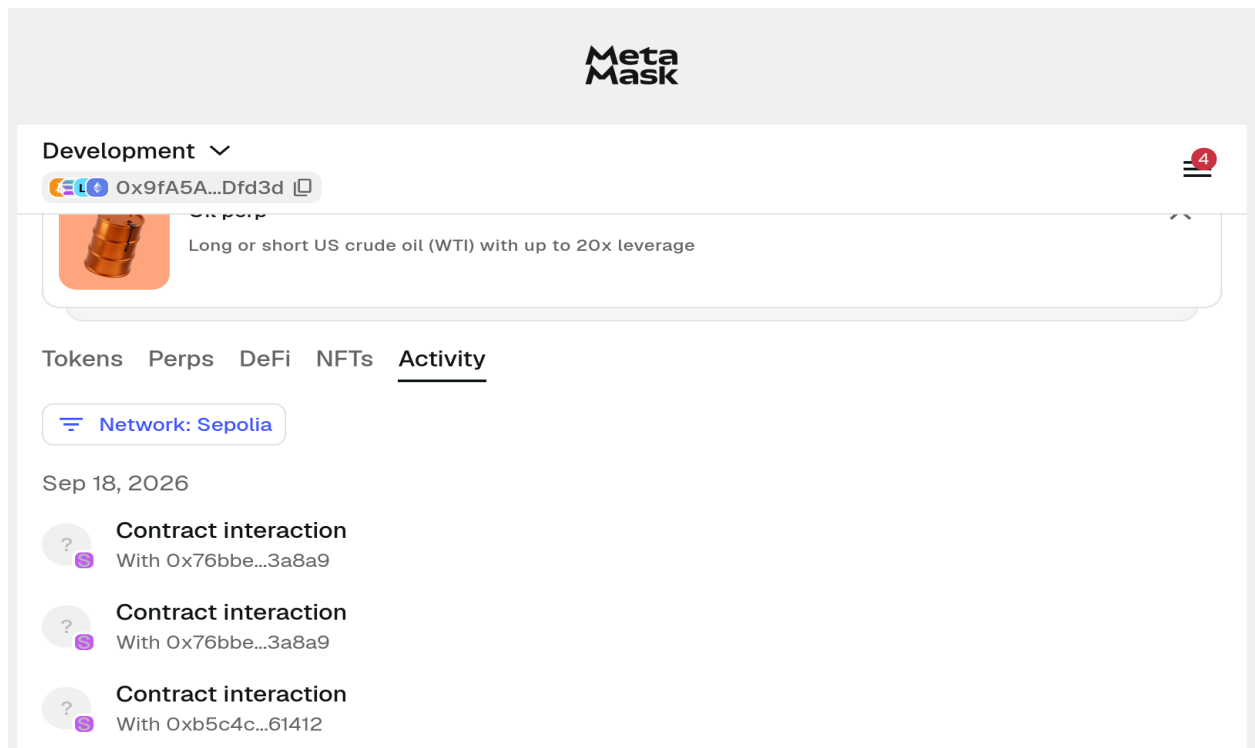


## Screenshot 6: MetaMask Transaction Confirmation





**Screenshot 7: Successful DApp Transaction**



**Screenshot 8: Transaction Verification**

Sepolia Testnet

Search by Address / Txn Hash / Block / Token

Overview

Logs (1)

State

</> API

TRANSACTION ACTION

0x9fA5A4c3...a99BDfd3d Call Submit Project Function on 0x76bbECaa...79593a8A9

[ This is a Sepolia Testnet transaction only ]

Transaction Hash:

0x7b4e79771f99b118378456a486f7268fb921fa9c0f67735bb6c1e328cc60c838

Status:

Success

Block:

11729680 31 Block Confirmations

Timestamp:

6 mins ago Sep-18-2026 08:58:24 AM +UTC

From:

0x9fA5A4c3386e333e520883b1649A0bca99BDfd3d

To:

0x76bbECaae5d377DE991a9e61670DF6A79593a8A9

Value:

0 ETH

Transaction Fee:

0.000560094606132633 ETH

Gas Price:

2.524301793 Gwei (0.000000002524301793 ETH)

## Conclusion

Thus, the MetaMask cryptocurrency wallet was successfully configured and integrated with the ZeroTrace DApp. The DApp was able to connect to the MetaMask wallet and initiate a blockchain transaction, which was approved by the user through MetaMask. The transaction was successfully recorded and verified using its transaction ID on the blockchain explorer. This experiment demonstrated the interaction between a cryptocurrency wallet, a decentralized application, smart contracts, and a blockchain network.